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SUPERSEDING
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FEDERAL SPECIFICATION

RING, LOOSE-LEAF BINDER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal loose-leaf binder rings with interlocking teeth.

1.2 Classification.

1.2.1 Type and grade. The loose-leaf binder rings shall be of one type and one grade.

1.2.2 Sizes. (See 3.3 and 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes; Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the Aermican Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.

3.1.1 Ring stock. Ring stock shall be round in cross section and of steel, having a hardness of not less than 65 on the Rockwell B scale when tested as described in 4.4.1.

3.1.2 Rivets. Rivets shall be of good commercial quality steel suitable for the intended purpose.

3.2 Design and construction. Rings shall be circular in shape and designed for easy opening and closing. The cross-sectional diameter of ring stock shall be as specified in table I. Each ring shall consist of two halves fastened together by hinged-joint construction with a rivet of suitable size. The rivet shall be upset to assure a firm hinged joint. Each ring half opposite the rivet shall have interlocking teeth designed to keep the ring closed securely.

3.3 Sizes. Ring sizes shall be as specified in table I. For the purposes of this specification size is defined as inside ring diameter. The dimensions of the ring are figured on the basis of the center part of the ring, left to right, with the riveted portion at the bottom, center.

TABLE I. Ring sizes and ring stock diameters

Ring sizes		Cross-sectional diameter of ring stock minimum
	Inch	Inch
3/4	+ 1/8, -1/16	0.099
1	+ 1/8, -1/16	.114
1-3/8	+ 3/16, -1/8	.114
2	± 1/8	.128
3	± 1/8	.181

3.4 Finish. Rings shall be bright finished and shall show no evidence of corrosion when tested as described in 4.4.2. Edges of cut, sheared, or drilled metal parts treated to resist corrosion before forming shall not be required to meet the test specified.

3.5 Workmamship. Rings shall be smooth without rough or sharp edges, burrs, or slivers. Rings shall be free from defects and imperfections which affect appearance, impair serviceability, or constitute a personal hazard.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all binder rings offered for inspection at one time, as defined in MIL-STD-105.

4.2.2 Sampling for visual and dimensional examination. A random sample of binder rings shall be taken from each lot in accordance with MIL-STD-105, inspection level S-3, acceptable quality level (AQL) 4.0 percent defective. The sample unit shall be one ring.

4.2.3 Sampling for test. A random sample of binder rings shall be taken from each lot in accordance with MIL-STD-105, inspection level S-1, acceptable quality level (AQL) 2.5 percent defective. The sample unit shall be one ring.

4.2.4 Sampling for preparation for delivery. Sampling shall be in accordance with MIL-STD-105. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Closure defects shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

4.3 Visual and dimensional examination. Samples taken in accordance with 4.2.2 shall be examined for defects listed in table II. Any rings in the sample containing one or more defects shall be rejected. If the number of defective rings in the samples exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE II. Classification of binder ring defects

Examine	Defects
Material	
Ring stock	Not round in cross section.
Rivets	Not suitable for intended purpose.
Design and construction	Not circular in shape. Do not open and close easily. Cross-sectional diameter or ring stock not as specified. Not made of two halves. Halves not fastened by hinged-joint construction. Rivet missing. Rivet not of suitable size. Rivet not upset Hinged joint not firm. Ring halves do not have interlocking teeth. Interlocking teeth do not keep ring closed securely.
Size	Ring size not as specified.
Finish	Not bright.
Workmanship	Not smooth. Evidence of rough or sharp edges. Evidence of burrs or slivers. Defects and imperfections which affect appearance, impair serviceability, or constitute a personal hazard.

4.4 Test procedures. Samples taken in accordance with 4.2.3 shall be subjected to the following tests. Any sample failing one or more tests shall be considered defective. If the number of defective rings in the sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.4.1 Hardness. Hardness shall be determined in accordance with ASTM E 18.

4.4.2 Corrosion resistance. Rings shall be tested for corrosion resistance by placing them in a boiling 10 percent (by weight) aqueous solution of sodium chloride for a period of 15 minutes. The rings upon being removed from this solution, shall be immediately immersed in 10 percent (by weight) aqueous solution of sodium chloride at room temperature. They shall then be removed from this solution and, without having the adhering liquid wiped off, allowed to dry for 24 hours at room temperature. The rings shall then be rinsed in lukewarm water, allowed to dry, and examined for evidence of corrosion.

4.5 Inspection of preparation for delivery requirements. Samples taken in accordance with 4.2.4 shall be examined to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping inadequate stapling. Bulging or distortion of container.

5. PREPARATION OF DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Rings in 3/4-inch thru 1-3/8-inch sizes shall be packaged in quantities of 100. Two-inch size rings shall be packaged in quantities of 50, and 3-inch size rings shall be packaged in quantities of 10. A close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676 shall be utilized as the unit container. Cushioning shall be provided. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B. The rings shall be packaged as specified in 5.1.1.1.

5.1.3 Level C. The rings shall be packaged in accordance with the suppliers commercial practice, providing that this will insure protection for the rings during shipment and provides for safe delivery to their destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Rings of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather resistant, grade V3c; or PPP-B-640, class 2. The fiberboard boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Unit packages of rings in 3/4-inch or 3-inch sizes shall be packed in quantities of 50; 1-inch or 1-3/8-inch ring sizes shall be packed in quantities of 25 and 2-inch size rings shall be packed in quantities of 20. A close-fitting box conforming to PPP-B-636, class domestic, style RSC, shall be utilized as the shipping container. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. The rings packaged as specified in 5.1 shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the National Motor Freight Classification rules and the Uniform Freight Classification rules.

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5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Rings covered by this specification are intended for use in securing perforated sheets and the like.

6.2 Ordering data. Purchasers shall exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of rings required (see 1.2.2 and 3.3).
(c) Levels of packaging and packing and any special marking required (see 5.1, 5.2, and 5.3).

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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